

MLFF-DATA6: Selection-Grade Structural and Foundation-Model Feature Specification

mdstats project

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1. Purpose and stage boundary

MLFF-DATA6 converts immutable DATA3 frame facts, DATA4 raw and optional profile-extension states, and DATA5 statistical roles into **raw selection evidence**. It introduces no fitted scaler, PCA basis, covariance metric, farthest-point selection, atomic reference fit, loss weight, MACE training file, or checkpoint decision. Those remain DATA7-DATA9 responsibilities.

The stage implements four contracts:

1. universal structural and optional profile-extension descriptors;
2. optional checkpoint-bound MACE invariant atomic descriptors;
3. DFT-label-derived difficulty residuals on authorized training domains only;
4. blinded prediction catalogs for non-training evidence domains.

The controlling information boundary is

geometry-only feature

may be computed on every role for which geometry access is authorized

label-derived residual

may be computed only on an authorized gradient-training domain

non-training prediction

may be stored without DFT residuals

locked test

remains sealed and unmaterialized before protocol freeze

1A. Current generic structural profile amendment

The historical sections below document the original DATA6 LTA and model-feature contracts. Beginning with 0.20.48a0, DATA6 also carries the material-neutral UniversalStructuralFeatureCatalog. Beginning with 0.20.49a0, DATA6 may carry one PhaseGeometrySelectionPlan derived from explicit DATA9A7a material contracts. Beginning with 0.20.53a0, DATA6 schema v5 may bind one completed, restartable checkpoint-bound model sweep.

The phase/geometry plan controls only MLFF selection exposure:

- enabled universal local-structure feature families;
- enabled generic geometry-event families;
- ordered atom-group coverage priorities for bulk, surface, interface, confined, and cluster geometries;
- advisory physical-observable call profiles;
- immutable plan/policy/material-contract lineage.

The numerical local-structure calculation remains owned by `mdstats.analysis.local_structure`. DATA6 does not own RDF, coordination- distribution, angle-distribution, topology, dynamics, transport, phonon, elastic, or thermodynamic algorithms.

DATA6-v1 through DATA6-v4 bundles remain readable. Historical evidence does not receive a fabricated phase/geometry plan or production model-sweep identity during deserialization.

2. Definitions

2.1 Raw descriptor

A raw descriptor is a deterministic numerical representation calculated without learning any dataset-dependent transformation. A MACE descriptor is an atomic feature after one or more message-passing blocks. MACE exposes these descriptors through `MACECalculator.get_descriptors()`; invariant-only output has one row per atom and a model-dependent feature dimension [1].

2.2 Difficulty feature

A difficulty feature compares a frozen foundation-model prediction with a DFT label. For frame i ,

$$e_{E,i} = \frac{|E_i^{\text{model}} - E_i^{\text{DFT}}|}{N_i},$$
$$e_{F,i} = \sqrt{\frac{1}{3N_i} \sum_{a=1}^{N_i} \|\mathbf{F}_{ia}^{\text{model}} - \mathbf{F}_{ia}^{\text{DFT}}\|_2^2},$$

and, when stress exists,

$$e_{\sigma,i} = \sqrt{\frac{1}{9} \|\boldsymbol{\sigma}_i^{\text{model}} - \boldsymbol{\sigma}_i^{\text{DFT}}\|_F^2}.$$

Species-resolved force errors are generated for every species present in the authorized frame. Profile-declared focus groups may later receive explicit objective or checkpoint emphasis, but DATA6 does not assume Li, Na, K, cations, or any material-specific species list. Difficulty features are selection evidence and must never be computed from outer-monitor, calibration, held-out-fold, or locked-test labels.

2.3 Blinded prediction

A blinded prediction contains model outputs or output summaries but no DFT residual, ranking, or label-derived selection field. Its record is bound to the DATA5 role and blinding boundary. A locked test produces a sealed metadata record only; prediction materialization is deferred until a protocol-freeze record exists.

3. Module ownership

`mdstats.analysis.local_structure`

universal material-neutral local-geometry calculations

`mdstats.training_data.structural_selection`

MLFF atom-group aggregation and universal selection evidence

`mdstats.training_data.lta_selection`

optional LTA extension implementation only

`mdstats.training_data.model_features`

checkpoint identities, optional MACE calculator adapter,
raw descriptor sidecars, and prediction summaries

`mdstats.training_data.difficulty`

authorized training domains, difficulty residuals,
blinded prediction domains and catalogs

`mdstats.training_data.production_model_sweep`

exact checkpoint-bound frame planning, sidecars, checkpoints, and resume

`mdstats.training_data.data6_bundle`

lineage-checked DATA6 orchestration

PyTorch and MACE are optional dependencies. Importing `mdstats` must not import MACE. The public calculator adapter imports MACE lazily only when a model-path constructor is requested.

4. Optional selection-grade LTA extension descriptors

4.1 Input requirements

The LTA builder requires:

- a DATA3 `TrainingFrameCatalog`;
- source-index-aligned `FrameData` for each run;
- a DATA4 `LtaPartitionFeatureCatalog`;
- DATA4 raw physical features.

DATA4 ring IDs and atom indices are treated as authoritative for this stage. Automatic natural-tiling discovery is not repeated in DATA6. A future profile may provide topology-derived ring IDs through the same record contract.

4.2 Per-cation environment vector

For every Li, Na, and K atom, the raw named vector contains:

- oxygen coordination;
- nearest, mean, standard-deviation, median, and maximum M-O distance;
- ring-center distance;
- signed and absolute ring-plane distance;
- radial off-center distance;
- one-hot ring-size indicators for 4R, 6R, 8R, and unresolved;
- one-hot site-class indicators;
- coordination-change, site-change, and ring-crossing flags.

Missing physical values are represented by a policy-declared finite fill value in the numeric named record and by an aligned missing mask. DATA7 decides how the mask and fill value enter a fitted metric; DATA6 does not impute from the dataset.

4.3 Per-frame LTA vector

The configuration-level vector contains:

- framework-integrity and event flags;
- counts by species, ring size, and site class;
- per-species mean, standard deviation, minimum, and maximum of the cation environment scalars;
- DATA4 framework pair-distance and coordination summaries;
- counts of unresolved cation environments.

Feature names are serialized beside values. Reordering a feature changes the policy digest and record digest.

5. Checkpoint-bound MACE descriptors

5.1 Checkpoint identity

Every descriptor and prediction record is bound to a `ModelCheckpointIdentity` containing:

```
model family
checkpoint locator
checkpoint SHA-256
calculator class
MACE version
adapter version
supported atomic numbers, when declared
device and default dtype
```

A descriptor produced by another checkpoint is a different feature block even when its shape is identical.

5.2 Calculator adapter

The MaceCalculatorProvider supports two constructors:

```
from_model_path(...)
    lazy-imports MACE and constructs MACECalculator

from_calculator(...)
    wraps an already constructed calculator and explicit checkpoint identity
```

The adapter calls `get_descriptors(atoms, invariants_only=..., num_layers=...)` and validates a finite two-dimensional array with one row per atom. Current MACE documentation states that only the base MACE and ScaleShiftMACE model classes support this descriptor path [1]. Unsupported calculators fail explicitly.

Predictions use ASE energy, force, and full 3×3 stress conventions. Missing stress is recorded rather than fabricated.

5.3 Array sidecars

High-dimensional descriptors are not embedded in JSON. Each frame is stored in one deterministic NumPy .npz file. In production, prediction outputs are stored independently as `predictions/<frame_uid>.npz`. The descriptor manifest records:

```
frame UID
relative path
shape
dtype
file SHA-256
array-content digest
```

The cache reader verifies the file hash, dtype, shape, finite values, and array-content digest before returning an array. Full provenance stays in JSON; raw arrays stay in sidecars.

5.4 Restartable production sweep

Data6ModelSweepPlan derives the exact descriptor and prediction frame sets from DATA5 roles and the active DATA6 policy. Locked-test, purged, excluded, and otherwise unauthorized frames are recorded as sealed or excluded and may not appear in the requested union.

Data6ModelSweepCheckpoint stores verified per-frame descriptor and prediction records with incomplete, complete, or failed status. Sidecars and checkpoint JSON are promoted atomically. Resume verifies file hashes, shapes, dtypes, finite values, content digests, and required-artifact completeness before reuse. A complete sweep provides a lazy persistent prediction cache so DATA6 residual and blinded-summary construction does not rerun the foundation model.

5.5 Native MACE descriptor autograd boundary

Native DATA6 graph batching separates descriptor extraction from prediction. Descriptor extraction is a forward-only operation and runs inside `torch.no_grad()`. Because MACE 0.3.16 defaults `compute_force=True`, the adapter must explicitly pass `False` for force, virial, stress, displacement, Hessian, edge-force, and atomic-stress outputs. A descriptor call must never enter `torch.autograd.grad`; doing so inside the no-gradient scope is a runtime error and wastes memory even when it succeeds outside that scope.

Prediction batching has a different contract. Energy, force, and optional stress are model outputs used by DATA6, so prediction executes inside `torch.enable_grad()` and retains MACE's derivative path. The adapter may not solve descriptor failures by globally disabling autograd or by suppressing the exception, because either action would invalidate force and stress predictions.

6. Authorized training domains

A TrainingDifficultyDomain is derived only from DATA5 assignments.

6.1 Final-development domain

Contains frames in the outer development role for one label domain. It is the only domain from which a final-training difficulty catalog may be built.

6.2 Cross-validation training domain

For fold k , contains only frames in `training_unit_ids`. Checkpoint-monitor, evaluation, and purge units are excluded. A fold-specific difficulty catalog is therefore valid input to fold-local DATA7 selection.

The builder verifies exact frame membership against the DATA5 unit catalog. A caller cannot create an arbitrary domain and label it as training evidence.

7. Blinded prediction domains

DATA6 creates domain metadata for:

- outer checkpoint monitor;
- uncertainty calibration;
- cross-validation checkpoint monitor;
- cross-validation held-out evaluation;
- locked interpolation test.

The first four may be materialized as prediction-only catalogs. The locked test is emitted with status `sealed_not_materialized` and zero prediction records. Purged and excluded roles produce no model feature catalog.

A blinded prediction record may contain predicted energy, force norms, species-resolved predicted-force norms, and predicted stress. It must not contain DFT energies, DFT forces, DFT stress, residuals, error ranks, or acceptance judgments.

8. DATA6 bundle

`Data6FeatureBundle` binds:

```
dataset ID
DATA2 source-catalog digest
DATA3 frame-catalog digest
DATA4 feature-bundle digest
DATA5 partition-bundle digest
universal structural and optional profile-extension catalogs
MACE raw-descriptor manifest, optional
atomic-model prediction manifest, optional
production model-sweep plan and checkpoint digest, optional
training-difficulty catalogs
blinded-prediction catalogs
checkpoint identity, when model features exist
```

The bundle fails when:

- any frame or unit lineage differs;
- a difficulty frame lies outside its authorized training domain;
- a blinded catalog contains residual fields;
- a locked-test catalog contains materialized predictions;
- a descriptor file does not verify;
- descriptors and predictions refer to different checkpoint identities;
- a production sweep does not realize the exact plan frame sets;
- a sweep plan, checkpoint, DATA5 bundle, DATA6 policy, or checkpoint identity disagrees.

9. Determinism and numerical policy

- Frame iteration follows sorted `frame_uid` order.
- Atomic rows retain source atom order.
- Feature names use a policy-defined stable order.
- Descriptor arrays use an explicit dtype.
- Sidecar filenames are `descriptors/<frame_uid>.npy`.
- JSON uses canonical sorted serialization and SHA-256 content digests.
- No random operation occurs in DATA6.

10. Public runtime records

The implemented public DATA6 contracts are:

```
UniversalStructuralFeatureCatalog
ProfileFeatureCatalog
ModelCheckpointIdentity
MaceDescriptorManifest
AtomicModelPredictionManifest
Data6ModelSweepPlan
Data6ModelSweepCheckpoint
TrainingDifficultyFeatureCatalog
BlindedEvaluationPredictionCatalog
Data6FeatureBundle
```

The real VASP-path acceptance tests use the supplied ASE 3.29.0 source distribution. ASE is a runtime dependency of frame materialization, while MACE and PyTorch remain optional lazy dependencies for checkpoint-backed features.

11. Tests and gate criteria

The stage gate requires:

1. exact round-trip serialization and tamper rejection;
2. Li/Na/K environment-vector fixtures with ring and coordination changes;
3. fixed feature-name and vector ordering;
4. fake-calculator descriptor extraction through the public MACE adapter;
5. descriptor sidecar hash, shape, and dtype verification;
6. final-development and fold-training domain membership checks;
7. rejection of residual computation on monitor, calibration, evaluation, or locked-test frames;
8. blinded catalogs containing no DFT residual fields;
9. locked tests remaining unmaterialized;
10. real ASE 3.29.0 VASP-to-DATA6 integration;
11. DATA0-DATA5 regression tests;
12. interruption, failure, resume, and corrupt-sidecar recovery;
13. DATA6 consumption of a completed sweep without repeated model calls;
14. a genuine MPA-0/Na-LTA descriptor and prediction smoke;
15. wheel/install/source-archive smoke tests.

12. Deferred work

DATA6 does not implement:

- robust scaling, PCA, whitening, or heterogeneous block metrics;
- farthest-point sampling or training-set size selection;
- fold-local or final atomic-reference energy fitting;
- training objectives and sample/property weights;
- complete production sweep execution;
- final/fold DATA7 production materialization;
- MACE replay binding, DATA8 job generation, or training execution;
- uncertainty calibration or active learning.

These remain DATA7-DATA10 responsibilities.

References

- [1] ACEsuit, “MACE descriptors,” MACE documentation, <https://mace-docs.readthedocs.io/en/latest/guide/descriptors.html>, accessed 2026-07-28.
- [2] I. Batatia et al., “MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields,” NeurIPS 2022, <https://arxiv.org/abs/2206.07697>.

[3] A. S. Rosen et al., “Machine learning the quantum-chemical properties of metal-organic frameworks for accelerated materials discovery,” *Matter* 4, 1578-1597 (2021). Descriptor diversity and model-error diagnostics are useful selection evidence, but they do not replace independent physical validation.

DATA9A7d amendment: optional selection-profile extensions

Beginning with 0.20.50a0, canonical DATA6 schema v4 stores optional material-specific selection evidence as `profile_selection_features`. The scientific payload remains owned by the extension provider. Generic DATA6 and DATA7 code may consume only the common frame-vector, atomic-environment, and environment-class adapters.

The previous `lta_selection_features` field is not emitted by new bundles. It remains a compatibility view and a v1-v3 deserialization input. The policy default no longer activates LTA construction; an explicit LTA profile extension or an explicitly requested historical compatibility path is required.

The generic fitted feature block is `profile_extensions`. `lta_frame` remains a read alias. Per-species raw, learned, and difficulty columns are derived from species present in the authorized fit domain and never from a fixed Li/Na/K list or from sealed roles.

DATA9A9a amendment: restartable checkpoint-bound model evidence

Beginning with 0.20.53a0, canonical DATA6 schema v5 may bind a complete `Data6ModelSweepPlan`, `MaceDescriptorManifest`, `AtomicModelPredictionManifest`, and sweep-checkpoint digest. The exact scientific plan is independent of how many bounded invocations are needed to finish it. Verified completed frames are reused; corrupt, incomplete, foreign, or policy-mismatched evidence fails or is explicitly recomputed according to the execution policy.

The production sweep does not activate locked-test predictions. It also does not complete DATA7, DATA8, replay binding, or DATA9B.